

Bhavya Kumar

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EDUCATION

University of Waterloo

Honours Bachelor of Mathematics (Co-op)

Waterloo, ON

Sept. 2025 – Present

- **Award:** President's Scholarship.
- **Relevant Coursework:** Data Types and Structures, Probability, Algorithm Design in C, Linear Algebra II, Calculus III, Functional Programming in Racket.

PROJECTS

Taskly - Kanban App with built-in AI Assistant - *React, Supabase, Vercel* | [Live Website](#) Apr. 2026

- Built and deployed a full-stack kanban application with multi-project boards, drag-and-drop via @dnd-kit, and a separate mobile component tree triggered by viewport detection for tab-based touch navigation.
- Designed a centralized mutation pipeline with undo/redo history and 500ms-debounced Supabase sync, maintaining a single source of truth across guest (localStorage) and authenticated modes with realtime cross-tab updates.
- Integrated Groq's DeepSeek-R1 70B across two distinct flows: full board generation from natural-language project descriptions, and command parsing with fuzzy task/column resolution and regex-fallback JSON extraction.
- Diagnosed a 60fps scroll regression caused by expensive CSS effects and resolved it with ref-based DOM mutations and passive event listeners; rebuilt the HTTP layer using Node.js native HTTPS with stream fallbacks to handle Vercel serverless timeouts.

Project Ghost - *Gemini 2.5 Flash, ElevenLabs, Python, Vultr* Jan. 2026

- Developed a multimodal code-analysis tool in a 2-person hackathon: Gemini 2.5 Flash simultaneously parses source code and UI screenshots to generate plain-English debugging reports, read aloud via ElevenLabs TTS.
- Wrote a Python backend orchestrating data flow between the Gemini vision model, the ElevenLabs pipeline, and the frontend; deployed the working prototype on Vultr for the demo.

Model Self-Driving Car Prototype - *Arduino, Embedded C/C++* 2023

- Built an Arduino car that avoids obstacles using ultrasonic sensors; wrote detection and motor control logic in C/C++ and tuned sensor thresholds through real-world testing for reliable navigation.

EXPERIENCE & LEADERSHIP

GDGI MUN

Head of IT

Apr. 2024 – Feb. 2025

- Directed IT logistics and media operations for a 2-day student-run Model UN conference, managing AV infrastructure, livestreaming, and registration systems across multiple committee rooms.

Clevered AI Fellowship

Student Fellow

Remote

Jan. 2024 – Feb. 2024

- Selected for a mentored AI program led by Oxford-affiliated researcher Ken Kahn; built a voice-activated accessibility prototype in Python using NLP-style intent matching to handle varied user commands.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, Racket, SQL

Frameworks & Libraries: React, Node.js, Express, Supabase, dnd-kit, REST APIs

Tools: Git, GitHub, Vercel, VS Code, Arduino IDE

ACHIEVEMENTS

Bronze Medalist: TIMO International Math Olympiad

Finalist: Northeastern University London Essay Competition (Tech & Society)

Author: "School of Machine Learning" — blog simplifying ML concepts for beginners